Authentication of a System using Audio File

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***Abstract*— Traditional text-based passwords are increasingly vulnerable to security threats such as brute-force attacks, keylogging, and shoulder surfing. Although graphical password systems have emerged to enhance security, they too are susceptible to observation-based attacks. To address these limitations, we propose a novel audio-based password authentication system that prioritizes both security and user convenience. In this system, users select specific click points within an audio file as their password, thereby eliminating visual cues and significantly reducing the risk of shoulder surfing. During registration, up to four audio click positions are chosen, encrypted using AES-128 and securely stored using Least Significant Bit (LSB) encoding. For successful authentication, users must reproduce the same sequence of click positions. To enhance accessibility and usability, the system integrates a forgot password feature .If a user fails to authenticate after three consecutive login attempts, the system automatically triggers a Time-based One-Time Password mechanism. An OTP is generated using the TOTP algorithm and sent to the user's registered email address. This OTP is valid for a limited time window and is securely stored on the server for verification. Upon successful OTP verification, the user is allowed to reset their password. This ensures a secure recovery process without compromising the system's integrity. Additionally, the system incorporates a virtual driver robot on the backend that assists users by automatically and securely inputting login credentials, enhancing usability while preserving data privacy. The use of persuasive security techniques encourages users to create less predictable audio patterns, thereby mitigating the risks of guessing and brute-force attacks. This multi-layered authentication system includes individual registration and login modules for platforms such as Facebook, Amazon, and the KCG portal, ensuring versatility and a seamless user experience across different environments.**

Keywords*— *Artificial Intelligence, Network Security, NLP(natural language processing) and Real time

I. Introduction

The evolution of Big Data and Cloud Computing has introduced both unprecedented opportunities and significant challenges, particularly in terms of security, scalability, and efficient resource management. As virtualization enables the dynamic allocation of virtual machines based on user demand, ensuring secure access to these resources becomes critical [11]. Traditional password-based authentication methods, though widely used, are increasingly vulnerable to threats such as key logging and password tracing. To address these concerns, this paper proposes an innovative audio-based authentication system that leverages sound localization and auditory recognition for enhanced security. Instead of relying on visual input or biometric data, the system prompts users to interact with specific points in an audio clip, making use of the human ability to recognize and recall sound patterns [2]. This interaction is then verified using audio positioning techniques, effectively mitigating the risk of common attacks like keystroke logging and shoulder surfing. The proposed method ensures both security and ease of use, offering a practical and efficient alternative to conventional authentication. Its integration with Big Data and Cloud environments further enhances its scalability and adaptability, making it a robust solution for modern digital platforms.

II. Ease of Use

.The proposed system introduces a Virtual AI Assistant that automates the login process by leveraging predefined music files as authentication keys. This AI-driven experience not only improves convenience but also enhances security and accessibility.

A. Virtual AI Assistant & Hands-Free Access

Automatically enters the registered email and password once the correct music file is detected. No need for manual input—just play the assigned music file for instant authentication.

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B. *Quick Response & NO memory Load*

Reduces the total login time to just a few seconds. Users don't have to remember or type complex passwords. Minimizes human errors such as mistyped credentials.

III. LITERATURE REVIEW

Rohan Rayarikar & Sanket Upadhyay (2012) – SMS Encryption using AES Algorithm on Android: presented an encryption scheme based on the AES algorithm to protect the communication of SMS on Android-based devices, promising secure data protection [1]. Naveen Kumar (2012) – User Authentication using Musical Password: Presented a music-based authentication method where users establish musical passwords stored as hashed sequences through the MD5 algorithm [2].

Smita Paira & Sourabh Chandra (2016) – Audio Cryptanalysis: She proposed an encryption technique by hybridizing symmetric key steganography for the secured transmission of message [3]. Nishith Sinha & B. Kishore (2015) – Encrypted Information Hiding using Audio Steganography and Audio Cryptography: A hybrid encryption scheme, which includes steganography of audio, text encryption, and transposition with the pseudo-random number generator, is proposed in [4].

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Srinivasan, S., & Raja, K. (2016). Preventing Cloud Attacks using Bio-Metric Authentication in Cloud Computing. Indian Journal of Science as well as technology. This research explores biometric authentication as a security measure to prevent unauthorized access in cloud computing. The proposed approach enhances data protection by integrating biometric features with cloud security mechanisms [14].

Muthuselvan S., Rajasekhar V., Shankar B., Mohana P., Doddabasappa N., Suresh B. (2023), Classification of Automatic Detection of Plant Disease in Leaves and Fruits using Image Processing with Convolutional Neural Network, ICOSEC 2023, explores CNN-based image processing for detecting plant diseases in leaves and fruits [15].

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IV. EXISTING SYSTEM

Graphical image password authentication involves users selecting specific points or regions within an image in a particular sequence as their password. These images are displayed through a graphical user interface (GUI), and the selected coordinates form the basis of authentication. While this method is considered more user-friendly than traditional text-based passwords, it presents several security concerns.

One of the primary vulnerabilities is its susceptibility to observation attacks. Users often select prominent or easily memorable areas within the image, making their choices predictable. These selections can be identified by others through simple observation or shoulder surfing, especially in public or shared spaces. Additionally, the limited variety of image hotspots increases the risk of successful guessing attacks. Despite offering a more visual and intuitive experience, the graphical image password system often compromises security due to human tendencies and predictable behavior patterns.

V.PROBLEM STATEMENT

Alphanumeric and graphical password-based authentication systems are vulnerable to security threats such as key logging, shoulder surfing, and password reuse. Users often reuse the same password across multiple accounts, leading to shadow attacks like Cross-Site Password Reuse (CSPR) and Intra-Site Password Reuse (ISPR). Graphical passwords also suffer from visibility issues, allowing attackers to observe the selection process. These challenges highlight the need for a more secure, efficient, and user-friendly authentication mechanism to reduce the risks of credential theft and visible input.

VI . PROPOSED SOLUTION / METHODOLOGY

The proposed solution introduces an audio-based authentication mechanism designed to improve both security and user convenience by leveraging sound position recognition, thereby eliminating visible input methods and effectively countering observation-based threats such as shoulder surfing and keylogging. During the registration phase, users select up to four specific click points within an audio file, which are encrypted using AES-128 and securely embedded through Least Significant Bit (LSB) encoding. For successful authentication, users must accurately reproduce the same sequence of audio click positions. To ensure accessibility, users are granted three login attempts, and if all attempts fail, A secure password recovery process is triggered via a Time-based One-Time Password (TOTP) sent to the user's registered email address. This allows them to safely reset their audio password within a limited time window (e.g., 5 minutes), ensuring both identity verification and protection against unauthorized access. Furthermore, to enhance usability and protect sensitive data, a virtual driver is integrated into the login process to automatically input credentials on behalf of the user, reducing input errors and minimizing exposure risk. The system also employs persuasive security techniques to encourage the creation of stronger, less predictable passwords and supports multi-platform authentication, ensuring seamless access across services such as Facebook, Amazon, and the KCG portal.

VII.MOTIVATION

The motivation behind this work is to enhance authentication security by moving away from traditional visual-based systems, which are susceptible to observation attacks such as shoulder surfing. This approach introduces an audio-based password mechanism**,** where encrypted click points are used to trigger the authentication process**.** The generated password is embedded into the audio file using steganography and securely stored**,** ensuring that sensitive information remains hidden from unauthorized access. During authentication, the user interacts with specific click positions during playback. When the correct positions are clicked, a decoding algorithm is triggered, successfully retrieving the hidden password. The system then authenticates the user and enables secure access to the site through a virtual driver. This method not only strengthens data protection but also provides a more discreet, accessible, and user-friendly alternative to conventional authentication techniques.This approach offers a more secure and user-friendly alternative to traditional and graphical password systems.

VIII. ALOGRITHM

AES (128-bit) Encryption is applied to secure the audio password because encrypted selected clicks stored or transmitted are not available for unauthorized access and confidentiality is maintained. LSB (Least Significant Bit) Coding is used for embedding the encrypted password in an audio file so that the embedded encrypted passwords cannot be retrieved or detected and, therefore enhance security through steganography.

A. AES – Advanced Encryption Algorithm

The Advanced Encryption Standard (AES) is one of the most widely used symmetric encryption algorithms, known for its efficiency and strong security. It requires minimal RAM space, making it suitable for devices with limited resources like smartphones and smart cards. AES is designed to perform faster due to its simplified operations at the hardware level, which allows quicker transitions compared to software-based processing. It supports key sizes of 128, 192, and 256 bits, all operating on a fixed 128-bit block size. The use of larger key sizes enhances security, making AES resistant to brute-force attacks and suitable for protecting sensitive data across various platforms.

B. LSB – Least Significant Bit technique

A widely used method in audio steganography is the Least Significant Bit (LSB) technique, which involves replacing the least significant bits of audio file bytes with the bits of the secret data to be hidden. This technique is effective because altering the LSB does not cause noticeable changes in audio quality, especially in high-resolution formats like 24-bit audio or images. In binary representation, the LSB is the bit at the rightmost position, holding the smallest value (1), and is often used to determine whether a number is even or odd. For example, in the binary number 01100001 (the ASCII representation of the letter 'a'), the rightmost bit is the LSB. To hide the letter 'a', the bits 01100001 can be embedded into the LSBs of eight bytes of a cover file. A decoding tool can then read these eight LSBs to extract and reconstruct the hidden message. This method is simple, efficient, and practical as

long as the modifications do not degrade the overall quality of the audio or image used as the cover medium.

C. Decryption and Authentication Mechanism

A decryption algorithm is employed during the login phase to enhance security. When the user clicks the correct point on the audio file, the system triggers the decryption process. This process involves extracting the encrypted password that was previously embedded into the audio file using steganography. If the extracted password matches the expected value, a virtual robot is initiated in the background to simulate the login process. The robot automatically inputs the verified password, granting the user secure access to the system. This approach not only ensures a high level of security but also automates the login process in a discreet and user-friendly manner.

D. Time-based One-Time Password

In this password recovery process, the Time-based One-Time Password (TOTP) algorithm is utilized to securely verify the user's identity after three failed login attempts. When the user selects "Forgot Password," a unique OTP is generated based on the current timestamp and a shared secret key, then sent to the user's registered email address. The OTP, typically 6-8 digits long, is valid for a short period to ensure timely and secure recovery. The user enters this OTP on the recovery page, and the system verifies it by generating the same OTP using the secret key and comparing it to the user-provided code. If the OTP is valid and within the allowed time frame, the user is granted access to reset their password, ensuring both security and usability in the recovery process.

IX.MODULE DESCRIPTION

Audio based authentication system is particularly designed for secure access and mainly for password invisibility. Where the encrypted message was concealed from the invader by being inserted into a carrier audio file.

Table 1.User Registration and Login Module

purpose	To Register User
Input	User-Name, Audio-Point Password, E-Mail, Gmail password, Phone No.
Processing	Registering into DB
Output	Registered in DB

The above table 1 describes about the User will be provided with a login page where he/she enters the credentials to complete the registration.

Table 2.Password Generation

s.no	Action	Description
1	Audio File Selection	User selects an audio file for authentication.
2	Audio Playback	The selected audio file plays in real-time.
3	User Clicks	The user clicks at specific moments while the audio is playing.
4	Position Identification	The system records the timestamps of up to four clicks.
5	Password Parameter Generation	The collected timestamps (audio signature) are consolidated into a unique password.
6	Encryption	The password parameters are encrypted using AES for security.

The above table 2 describes about the User selects the audio file and points the position (Audio Signature) while the audio file is running and the position of the clicks is identified. Users can only make four clicks then the consolidated points becomes as a password parameter.

Table 3.Password Embedding

S.NO	Action	Description
1	Password Generation	User-generated audio signature(click positions) is consolidated into a password.
2	LSB Encoding	The encrypted password is embedded into the original using the Least Significant Bit (LSB) in an audio file.
3	Audio Password Storage	The modified audio file containing the embedded password is stored securely in the MySQL database.

The above table 3 describes about the audio password gets encrypted using AES algorithm (128bit) and the encrypted password gets embedded using the LSB coding method in the original audio file. The generated audio password file gets stored in the MySQL database.

Table 4.Authentication

Purpose	To Authenticate User.
Input	User-Name, Audio-Point Password.
Authentication Retrieval	During the login process, the system retrieves the stored audio file and applies a reverse steganographic technique to extract and decrypt the embedded password. This decrypted password is then verified for authentication.
Output	User Authenticated and login into selected website.

The above table 4 describes about the authentication. If user is (valid) - During the authentication phase, if the audio-clicks are made exactly it matches the

parameter in the database, then the user gets authenticated else rejected.
If user is (invalid) - Here the login attempts are limited to three times then the User ID gets locked. The User ID becomes invalid until the authorized user changed the password through password reset option.

XI . Comparison Graph Description

A bar chart can be used to visually compare key aspects such as Security Level, User Convenience, Encryption Strength, and Vulnerability to Attacks. The x-axis represents the features, while The rating scale is represented by the y-axis (e.g., 1-10).

A. **Vulnerability to Attack**: Graphical password systems are highly visible and thus more prone to observation attacks. In contrast, the audio-click authentication approach makes it difficult for attackers to identify input points due to the absence of visual cues.

B. **Scalability**: The new method is designed to function seamlessly across multiple platforms such as Facebook, Amazon, and the KCG portal, ensuring broad applicability.

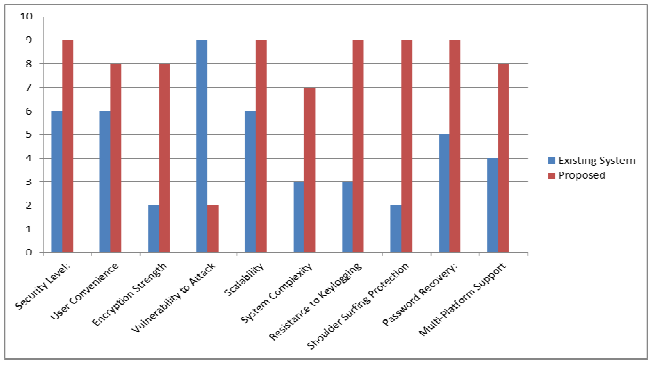
C. **System Complexity**: While the audio-based mechanism introduces additional backend components like encryption and virtual drivers, it remains manageable and efficient from the user's perspective.

D. **Resistance to Key logging**: Since the proposed approach eliminates the need for typing, it offers strong protection against key loggers that rely on capturing keystrokes.

E. **Shoulder Surfing Protection**: By shifting from visual interaction to audio cues, the advanced system significantly reduces the risk of shoulder surfing.

F. **Password Recovery**: In case of failed login attempts, a secure OTP-based recovery process helps users reset their credentials through their registered email, ensuring both security and convenience.

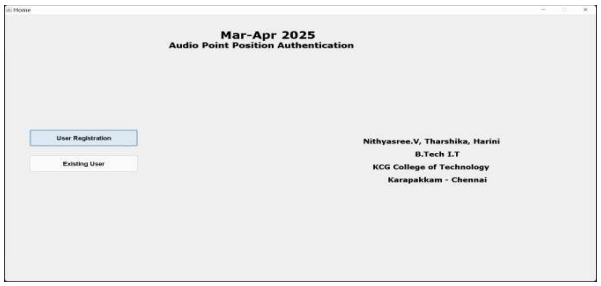
G. **Multi-Platform Support**: Through the integration of a virtual driver, this solution enables automatic and secure login across a variety of services, enhancing user accessibility.



X. Result & Analysis

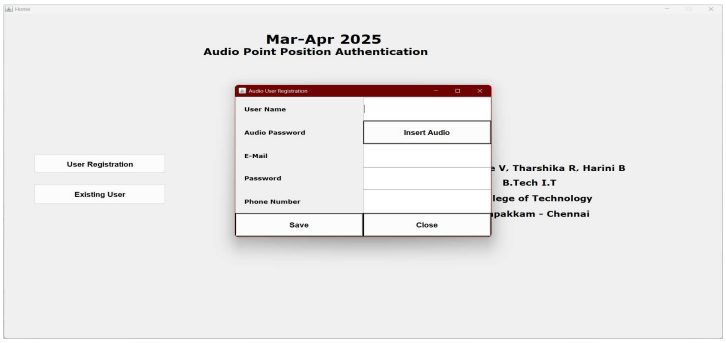
The system ensures secure login with audio click passwords, encrypts them using AES-128 and LSB, verifies user input, and prevents attacks with a decoy file and a secure virtual driver.

Figure 1.Initial page for authentication using audio files



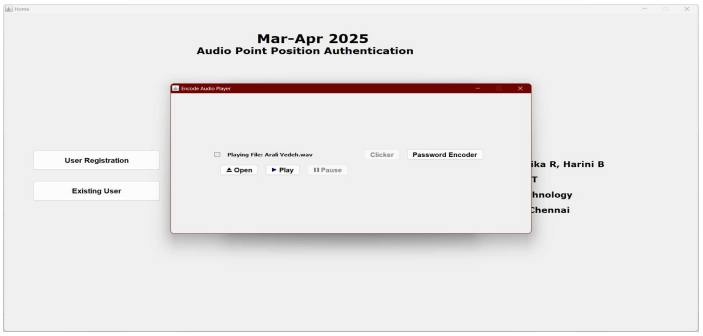
The above Figure 1 describes about the home page includes User Registration for new users to register an account with required information and Existing User Login for previous users to log in to their accounts without any hassle. It provides a hassle-free onboarding process with validation and secure authentication through the assistance of virtual robot.

Figure 2.user registration page for authentication using audio files



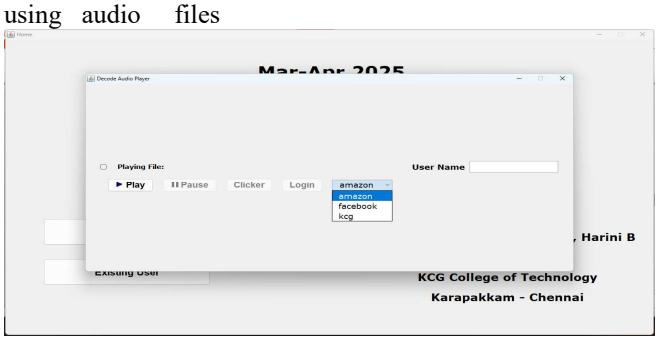
The above Figure 2 describes about User registration begins when a registered user wants to sign up for secure authentication. The user enters necessary details such as username, email, and phone number. They then record and set an audio password by selecting specific click points in an audio file. This password is encrypted using the AES-128 algorithm and embedded using LSB encoding. The encrypted password is securely stored in the database. Upon completion, the user is successfully registered.

Figure 3.Encoded page for authentication using audio files



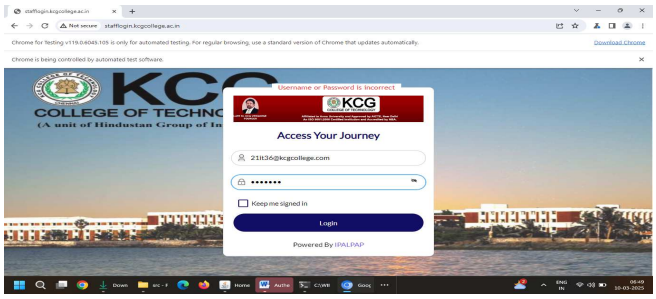
The Figure 3 illustrates the Encode Audio Player, which allows users to select four specific click points in an audio file. Passwords from platforms like Amazon, Facebook, and the KCG College portal are embedded into the audio using steganography. The resulting stego-audio file is securely stored in the database. This file serves as the authentication key for future logins.

Figure 4.decoded audio or login page for authentication using audio files



The above Figure 4 describes about the A registered user initiates authentication using the audio-based system by entering their username. The system plays the stored stego-audio file, and the user must accurately reproduce the predefined click sequence. The system then decrypts and verifies the click positions. If the sequence matches, the user is successfully authenticated and granted access. If incorrect, a decoy audio file is played to mislead potential attackers.

Figure 5.Virtual driver page for authentication using audio files



The above Figure 5 illustrates the AI Virtual Drive, a secure system that automatically creates unique login credentials for the Facebook login page. These credentials are protected using encryption and steganography, ensuring they remain hidden from both users and potential attackers. The virtual drive enables a seamless and highly secure login process, effectively preventing unauthorized access and maintaining data confidentiality.

XII.Result & Discussion

During authentication, the system verifies access by matching the user's selected positions with predefined click points. If authentication fails, access is denied. To enhance security, login attempts are limited to three. After three failed attempts, the system blocks further access to prevent unauthorized entry and brute-force attacks. Experimental results show this method improves security while maintaining user accessibility.

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XIII. Conclusion

The proposed approach uses two algorithms for secure authentication. The first algorithm applies quantization of audio point positions in the transform domain and uses AES encryption to embed secure data within the domain coefficients. The second algorithm uses a reversible integer transform, where the secure key is added as an extra bit after analyzing the binary representation of the integer coefficients using LSB coding.

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